

Nikhil Raghav

PhD Student in Computer Science

AIML Group, Institute for Advancing Intelligence, TCG CREST, Salt Lake, Kolkata, India

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Summary

I am currently a senior research fellow (SRF) at the Institute for Advancing Intelligence (IAI), TCG CREST, Kolkata, working under the guidance of Prof. Md Sahidullah. My PhD is jointly affiliated with the Department of Computer Science, RKMVERI University, Belur Campus, where I am guided by Swami Punyeshwarananda (earlier life name Prof. Vikas Reddy), Head of Department (HOD).

Research Interests

Speaker diarization (offline and streaming), deep and classical clustering techniques, graph-based learning for speech applications, representation learning, and speech foundation models.

Education

Institute for Advancing Intelligence, TCG CREST, Kolkata, India

PhD in Computer Science

August 2022 – Present

Thesis- *Advances in Speaker Diarization*

Indian Institute of Technology (IIT) Mandi, Himachal Pradesh, India

Master of Science in Applied Mathematics

December 2019 – August 2017

Advisor: Associate Prof. Sarita Azad

Thesis - Quantifying the changes in temperature and precipitation over the North-West Himalayan region.

Dyal Singh College, University of Delhi, New Delhi, India

Bachelor of Science with Honours in Mathematics,

June 2016 – August 2013

Honors & Awards

INTERNATIONAL Fellowship

2024 **Linguistic Data Consortium(LDC), Data Scholarship**, University of Pennsylvania, USA.

2018 **DAAD-Fellowship**, Germany

NATIONAL Fellowship

2022 **Institute Doctoral Fellowship**, TCG CREST

2018 **Merit Cum Means Fellowship**, IIT Mandi

National Exams

2022 Qualified GATE in Mathematics (MA)

2017,16 Qualified IIT JAM in Mathematics (MA)

Publications

Conferences

1. Nikhil Raghav, Avisek Gupta, Md Sahidullah, and Swagatam Das, *Self-Tuning Spectral Clustering for Speaker Diarization*, **IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)**, Hyderabad, India, 6-11 April 2025.
2. Tanmoy Jana, Nikhil Raghav, Angshul Majumdar, and Md Sahidullah, *Exact Rotation Invariant Robust PCA*, **IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)**, Hyderabad, India, 6-11 April 2025.
3. Nikhil Raghav and Md Sahidullah, *Assessing the Robustness of Spectral Clustering for Deep Speaker Diarization*, **IEEE India Conference (INDICON)**, IIT Kharagpur, India, 19-21 Dec 2024.

In Preparation

1. Nikhil Raghav, and Md Sahidullah, *Behind the Scenes: How Dataset Factors Shapes Speaker Diarization*, to be communicated to the Journal of the Acoustical Society of America(JASA).
2. Nikhil Raghav, Avisek Gupta, Swami Punyeshwarananda, Swagatam Das, and Md Sahidullah, *Self-Tuning Spectral Clustering with Adaptive Neighbor Selection for Speaker Diarization*, to be communicated to IEEE Transactions on Audio, Speech and Language Processing.
3. Tanmoy Jana, Nikhil Raghav, Angshul Majumdar, and Md Sahidullah, *Mean-Corrected R0-PCA with Applications in Data Representation and Analysis* to be communicated to IEEE Transactions on Pattern Analysis and Machine Intelligence.

Workshops & poster presentations

- ◇ Summer School on Speech Signal Processing (S4P) 2024, organized by: DA-IICT, Gandhinagar, Gujarat, from 6-10 July 2024.
- ◇ Winter School on Speech And Audio Processing (WiSSAP-2024), organized jointly by IIT Madras and KL University, Vaddeswaram, Andhra Pradesh, from 9-12 December 2024.

Peer Review Activities

- ◇ IEEE Transactions on Multimedia
- ◇ Computer Speech & Language

Technical Skills

- ◇ *Programming Languages*: Python, C, C++, Shell
- ◇ *ML Libraries*: PyTorch, Scikit-learn, HuggingFace Transformers
- ◇ *ASR Frameworks*: Kaldi (basic), SpeechBrain, pyannote-audio
- ◇ *Software & Tools*: Audacity, Git, Jupyter, Colab, SQL, Visual Studio Code
- ◇ *Document Preparation Systems*: LaTeX, Google Suite (Docs, Sheets, Slides)
- ◇ *Platforms*: Kaggle, GitHub, Stack Overflow
- ◇ *Operating Systems*: Linux, macOS, Windows

Leadership Activities

- ◇ Session Chair, Cybersecurity — Information Forensics & Security Track, ICASSP 2025

Community Service

- ◇ Educational Content Creator on YouTube: Created and managed the “*VigyanBharat*” channel, providing free educational content on Mathematics to aid in learning.
YouTube Link: <https://www.youtube.com/@VigyanBharat>
- ◇ Organised a workshop titled *AI for everyone: Tools and strategies for Productivity, Advancing Careers, and Mathematical Insight* at Jawaharlal Nehru College, Pasighat, Arunachal Pradesh from 18-19 March 2025 for the bachelor’s students in Mathematics.

Memberships in Professional Associations

- ◇ IEEE Student Member since 2024 (Membership No. 100173868)

Personal Skills & Competences

- ◇ *Language Proficiencies*: English (Fluent), Hindi (Fluent), French (Basic proficiency), German (A1)
- ◇ *Hobbies*: Yoga, Badminton, Football, Trekking

References

- ◇ *Dr. Md Sahidullah*, Assistant Professor, Institute for Advancing Intelligence(IAI), TCG CREST, Kolkata, md.sahidullah@tcgcrest.org, (Doctoral Advisor)
- ◇ *Swami Punyeshwarananda*, Assistant Professor & HOD, Department of Computer Science, RK-MVERI University, Belur Campus, punyeshwarananda@gm.rkmvu.ac.in, (Doctoral Advisor)
- ◇ *Dr. Swagatam Das*, Professor & Head Electronics and Communication Sciences Unit, Indian Statistical Institute, Kolkata, swagatam.das@isical.ac.in (Mentor, AIML Group, TCG CREST)
- ◇ *Dr. Angshul Majumdar*, Professor, Institute for Advancing Intelligence(IAI), TCG CREST, Kolkata, angshul.majumdar@tcgcrest.org

Contact

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